

EXPERIMENT 2

Aim: Create a Blockchain using Python

Theory:

Blockchain is a distributed and immutable ledger that records data in a sequence of linked blocks. Each block contains a timestamp, proof (from Proof-of-Work), and the hash of the previous block, ensuring integrity and security.

In this program:

1. **Block Creation** – A block is generated using `create_block()` with a proof and previous hash.
2. **Proof-of-Work (PoW)** – Implemented to mine a block by solving a cryptographic puzzle (finding a hash with leading zeros).
3. **Hashing** – SHA-256 algorithm ensures data immutability.
4. **Validation** – `is_chain_valid()` checks that every block correctly references the previous hash and satisfies PoW conditions.
5. **Flask Web API** – Routes `/mine_block`, `/get_chain`, and `/is_valid` allow mining, fetching the chain, and verifying blockchain integrity through a browser or API client.

This program runs on a local server and simulates mining a blockchain without a network of nodes, making it an ideal introductory model to understand blockchain basics.

Program and Output:

Program:

```
import datetime
import hashlib
import json
```

```

class Blockchain:
    def __init__(self):
        self.chain = []
        self.transactions = []
        self.create_block(proof=1, previous_hash='0')

    def create_block(self, proof, previous_hash, transactions=None):
        block = {
            'index': len(self.chain) + 1,
            'timestamp': str(datetime.datetime.now()),
            'proof': proof,
            'previous_hash': previous_hash,
            'transactions': transactions if transactions is not None else []
        }

        self.chain.append(block)
        return block

    def get_previous_block(self):
        return self.chain[-1]

    def create_Transactions(self, sender, receiver, amount):
        transaction = {
            'sender': sender,
            'receiver': receiver,
            'amount': amount
        }

        self.transactions.append(transaction)

        return transaction

    def proof_of_work(self, previous_proof):
        new_proof = 1

```

```
while True:
```

```
    hash_operation = hashlib.sha256(  
        str(new_proof**2 -  
            previous_proof**2).encode()  
    ).hexdigest()
```

```
    if hash_operation[:3] == '000':
```

```
        return new_proof
```

```
    new_proof += 1
```

```
def hash(self, block):
```

```
    encoded_block = json.dumps(  
        block,
```

```
        sort_keys=True
```

```
    ).encode()
```

```
    return hashlib.sha256(encoded_block).hexdigest()
```

```
def is_chain_valid(self, chain):
```

```
    previous_block = chain[0]
```

```
    block_index = 1
```

```
    while block_index < len(chain):
```

```
        block = chain[block_index]
```

```
        # Check previous hash
```

```
        if block['previous_hash'] != self.hash(previous_block):
```

```
            return False
```

```
        # Check Proof of Work
```

```
        previous_proof = previous_block['proof']
```

```
        proof = block['proof']
```

```
hash_operation = hashlib.sha256(
    str(proof**2 - previous_proof**2).encode()
).hexdigest()
```

```
if hash_operation[:3] != '000':
    return False
```

```
previous_block = block
block_index += 1
```

```
return True
```

```
# Creating a Blockchain
```

```
blockchain = Blockchain()
```

```
print(blockchain.chain)
```

Output:

```
[{'index': 1, 'timestamp': '2026-08-30 15:28:43.662032', 'proof': 1, 'previous_hash': '0',
'transactions': []}]
```

Program:

```
# Mining a new block
```

```
def mine_block():
```

```
    # Check whether transactions are available
```

```
    if len(blockchain.transactions) == 0:
```

```
        return {
```

```
            'message': 'No transactions available. Block cannot be mined.'
```

```
        }
```

```
    previous_block = blockchain.get_previous_block()
```

```
    previous_proof = previous_block['proof']
```

```
    proof = blockchain.proof_of_work(previous_proof)
```

```

previous_hash = blockchain.hash(previous_block)

# Store current transactions in the block
transactions = blockchain.transactions.copy()

# Remove transactions before mining the block
blockchain.transactions.clear()

block = blockchain.create_block(
    proof,
    previous_hash,
    transactions
)

response = {
    'message': 'Congratulations, you just mined a block!',
    'index': block['index'],
    'timestamp': block['timestamp'],
    'proof': block['proof'],
    'previous_hash':
        block['previous_hash'], 'transactions':
        block['transactions']
}

return response

sender = input("Enter sender: ")
receiver = input("Enter receiver: ")
amount = input("Enter amount: ")

transaction = blockchain.create_Transactions(
    sender,
    receiver,
    amount
)

```

```
print("Transaction added successfully:")
print(transaction)
```

Output:

```
Enter sender: janhavi
Enter receiver: bob
Enter amount: 40000
Transaction added successfully:
{'sender': 'janhavi', 'receiver': 'bob', 'amount': '40000'}
```

Program:

```
def get_chain():
    response = {
        'chain': blockchain.chain,
        'length': len(blockchain.chain),
        'pending_transactions': blockchain.transactions
    }

    return response

def is_valid():

    valid = blockchain.is_chain_valid(blockchain.chain)

    if valid:
        response = {
            'message': 'All good. The Blockchain is valid.'
        }
    else:
        response = {
            'message': 'Houston, we have a problem. The Blockchain is not valid.'
        }

    return response
```

```
print("Functions Menu:")
print("=====")
print("1. Add a transaction")
print("2. Mine a block")
print("3. Display the chain")
print("4. Check the validity of the chain")
```

```
choice = int(input("Enter your choice: "))
```

```
if choice == 1:
```

```
    sender = input("Enter sender: ")
```

```
    receiver = input("Enter receiver: ")
```

```
    amount = input("Enter amount: ")
```

```
    print(
```

```
        blockchain.create_Transactions(
```

```
            sender,
```

```
            receiver,
```

```
            amount
```

```
        )
```

```
    )
```

```
elif choice == 2:
```

```
    print(mine_block())
```

```
elif choice == 3:
```

```
    print(get_chain())
```

```
elif choice == 4:
```

```
    print(is_valid())
```

else:

```
print("Invalid choice")
```

Output:

Functions Menu:

=====

1. Add a transaction
 2. Mine a block
 3. Display the chain
 4. Check the validity of the chain
- Enter your choice: 3

```
{'chain': [{'index': 1, 'timestamp': '2026-08-30 15:28:43.662032', 'proof': 1, 'previous_hash': '0', 'transactions': []}, {'index': 2, 'timestamp': '2026-08-30 15:29:49.789595', 'proof': 533, 'previous_hash': 'a779f0fec2265dbd9497ba53771b68cc911847b9b9415dbf19e07f77a2f56300', 'transactions': [{'sender': 'janhavi', 'receiver': 'bob', 'amount': '40000'}]}, {'index': 3, 'timestamp': '2026-08-30 15:30:55.901101', 'proof': 1066, 'previous_hash': '5400000000000000000000000000000000000000000000000000000000000000', 'transactions': []}], 'length': 3, 'pending_transactions': []}
```

Functions Menu:

=====

1. Add a transaction
 2. Mine a block
 3. Display the chain
 4. Check the validity of the chain
- Enter your choice: 4
- ```
{'message': 'All good. The Blockchain is valid.'}
```



## Output (Postman):

```
1 {
2 "message": "Transaction added successfully.",
3 "transaction": {
4 "amount": 40000,
5 "receiver": "siu",
6 "sender": "janhavi"
7 }
8 }
9 |
```

```
1 {
2 "message": "Transaction added successfully.",
3 "transaction": {
4 "amount": 40000,
5 "receiver": "siu",
6 "sender": "janhavi"
7 }
8 }
9 |
```

```
1 {
2 "index": 2,
3 "message": "Congratulations, you just mined a block!",
4 "previous_hash": "80f7f57b4eb6dc9033f4b9102dd5f97f8afb7624f9b397ac80ce0510b03219f9",
5 "proof": 533,
6 "timestamp": "2026-08-30 21:31:34.900695",
7 "transactions": [
8 {
9 "amount": 40000,
10 "receiver": "riyu",
11 "sender": "janhavi"
12 }
13]
14 }
15
```

```

1 {
2 "chain": [
3 {
4 "index": 1,
5 "previous_hash": "0",
6 "proof": 1,
7 "timestamp": "2026-08-30 21:19:26.794974",
8 "transactions": []
9 },
10 {
11 "index": 2,
12 "previous_hash": "80f7f57b4eb6dc9033f4b9102dd5f97f8afb7624f9b397ac80ce0510b03219f9",
13 "proof": 533,
14 "timestamp": "2026-08-30 21:31:34.900695",
15 "transactions": [
16 {
17 "amount": 40000,
18 "receiver": "riyu",
19 "sender": "janhavi"
20 }
21]
22 }
23],
24 "length": 2,
25 "pending_transactions": []
26 }
27

```

```

1 {
2 "message": "All good. The Blockchain is valid."
3 }
4

```

## Conclusion:

We successfully created a basic blockchain using Python and Flask that supports block mining, chain retrieval, and validity checks. Proof-of-Work ensures security by making block creation computationally intensive, while cryptographic hashing guarantees immutability. The implementation demonstrates core blockchain principles—decentralization, immutability, and consensus—in a simplified environment, providing a strong foundation for building more advanced blockchain systems.